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Tomás S. R. Silva

Curriculum Vitæ

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Education

PhD, Mathematics — Aug 2022 – Feb 2027 (expected)
University of Campinas (UNICAMP) — Campinas, São Paulo, Brazil
Supervisor: Prof. Henrique N. Sá Earp

BSc, Computer Engineering — Mar 2017 – Dec 2021
University of Campinas (UNICAMP) — Campinas, São Paulo, Brazil
Supervisor: Prof. Ricardo Dahab

Research Experience

Visiting Research Student — Jan 2024 – Jul 2024
City, University of London — London, UK
Supervisor: Prof. Yang-Hui He (London Institute for Mathematical Sciences & Merton College, Oxford)

PhD Researcher — Aug 2022 – present
Laboratory of Gauge Theory and Algebraic Geometry –IMECC/UNICAMP;
and GeometrIA LAB – Brazilian Center for Geometry

Junior Researcher — May 2018 – Dec 2021
Security and Cryptography Laboratory (LASCA), UNICAMP

Publications

• Peer-Reviewed Journal Articles

The exterior derivative and the mean value equality in $\mathbb{R}^n$ — 2026
D. Fadel, H. N. Sá Earp & T. Silva
arXiv:2510.00999 — *São Paulo Journal of Mathematical Sciences, Vol. 20, 2026*.

Neural and numerical methods for G_2-structures on contact Calabi–Yau 7-manifolds — 2026
E. Heyes, E. Hirst, H. N. Sá Earp & T. Silva
Physics Letters B, Vol. 878, 2026. ◇ arXiv:2602.12438

Metaheuristic Generation of Brane Tilings — 2025
Y.-H. He, V. Jejjala & T. Silva
Physics Letters B, Vol. 862, 2025. ◇ arXiv:2412.19313

Efficient isochronous fixed-weight sampling with applications to NTRU — 2024
D. L. Gazzoni Filho, T. Silva & J. López
IACR Communications in Cryptology, Vol. 1, No. 2, 2024.

Machine-learning Sasakian and G_2 topology on contact Calabi–Yau 7-manifolds — 2024
D. Aggarwal, Y.-H. He, E. Heyes, E. Hirst, H. N. Sá Earp & T. Silva
Physics Letters B, Vol. 850, 2024. ◇ arXiv:2310.03064

• Peer-Reviewed Conference Proceedings

A data-driven view of exterior calculus — 2026
T. Silva
AI in Mathematics and Theoretical Physics

Machine learning topology of Calabi–Yau links — 2025
T. Silva & H. N. Sá Earp
Proceeding Series of the Brazilian Society of Computational and Applied Mathematics, Vol. 11, No. 1, 2025.

• Preprints

Minimising Willmore Energy via Neural Flow — 2026
E. Hirst, H. N. Sá Earp & T. Silva
arXiv:2604.04321

A penalised Saito functional for heuristic search of free line arrangements — 2026
T. Silva
arXiv:2604.02995

Honours & Awards

Academic Distinction 2022

Awarded by the Institute of Computing, UNICAMP, for academic excellence during the undergraduate programme.

Best Undergraduate Research Project 2021

Awarded by the Institute of Computing, UNICAMP, for the work *"Characterizing Lattices for Cryptography"*, recognised as a significant contribution to scientific and technological research in Computing that year.

Invited Talks

Machine learning G_2 Geometry 2026

Exceptional Geometry in Lyon, Camille Jordan Institute &
14th International Conference on Clifford Algebras and Their Applications in Mathematical Physics, University of Erfurt

Digital Tools and AI for Contemporary Mathematical Productivity 2026

Brazilian Center for Geometry – CBG.

Mathematics as a game: exploring freeness of line arrangements with machine learning 2026

Mathematics Department Seminar, Institute of Mathematical and Computer Sciences (ICMC), University of São Paulo (USP).

A data-driven view of exterior calculus 2026

DANGER: Data, Numbers, and Geometry — Banff International Research Station (BIRS).

From Theory to Computation: Unravelling Hyperplane Arrangements 2025

Geometry, Algebra and Combinatorics Seminar, UNICAMP.

Machine Learning Ricci-Flat Metrics 2025

São Paulo Meeting on Geometry, Institute of Mathematics and Statistics (IME), University of São Paulo (USP) & Geometry Seminar, Institute of Mathematical and Computer Sciences (ICMC), University of São Paulo (USP) &
AI+Math Seminar, Brazilian Institute of Data Science (BI0S).

Machine Learning Riemannian Metrics with Special Holonomy 2025

Machine Learning and Mathematics programme, Korea Institute for Advanced Study (KIAS).

From Theory to Computation: Unravelling Hyperplane Arrangements 2024

Seminar on Commutative Algebra and Algebraic Geometry, Federal University of Pernambuco (UFPE).

Learning Geometry and Topology of Algebraic Varieties 2023

AI+Math Seminar, Brazilian Institute of Data Science (BI0S).

Grants & Fellowships

• Proposal Title: Geometry, Topology and Data Science
Funding Agency: São Paulo Research Foundation (FAPESP), grant no. 2022/09891-4,
within the FAPESP–ANR Thematic Project *BRIDGES: France–Brazil Interactions in Gauge Theory, Extremal Structures and Stability*, grant no. 2021/04065-6
Role: Doctoral fellow | Principal Investigator: Prof. Henrique N. Sá Earp
Coverage Period: Sep 2022 – present

• Proposal Title: Post-Quantum Cryptography
Funding Agency: Samsung R&D Institute Brazil
Role: Undergraduate research grantee | Principal Investigators: Prof. Ricardo Dahab & Prof. Julio López
Coverage Period: Nov 2020 – Nov 2021

• Proposal Title: Characterizing Lattices and Codes for Cryptography
Funding Agency: São Paulo Research Foundation (FAPESP), grant no. 2019/19296-3
Role: Undergraduate research grantee | Principal Investigator: Prof. Ricardo Dahab
Coverage Period: Nov 2019 – Oct 2020

• Proposal Title: Characterizing Lattices for Cryptography
Funding Agencies: São Paulo Research Foundation (FAPESP), grant no. 2018/08130-4; and National Council for Scientific and Technological Development (CNPq), grant no. 124431/2019-4
Role: Undergraduate research grantee | Principal Investigator: Prof. Ricardo Dahab
Coverage Period: May 2018 – Apr 2019 (FAPESP) & Aug 2019 – Nov 2019 (CNPq)

Teaching & Advising

• **Courses — Teaching Assistant (UNICAMP)**

AI for Geometry ..... Aug 2026 – Sep 2026
Linear Algebra ..... Aug 2025 – Dec 2025 & Feb 2026 – Jul 2026
Vector Calculus (Calculus II) ..... Feb 2025 – Jul 2025 & Feb 2026 – Jul 2026
Analytic Geometry ..... Aug 2022 – Dec 2022 & Aug 2023 – Dec 2023

• **Undergraduate Student Co-Supervision**

Bianca Bosnardo April 2026 – March 2027
Undergrad research project: *"An introduction to hyperplane arrangements".*

Service & Professional Development

• **Service to the Discipline and Profession**

Founder and Moderator — MathIA AmLat 2026–present
Founder and coordinator of a Latin American community dedicated to discussion and collaboration on applications of artificial intelligence in mathematical research. math-ia-amlat.zulipchat.com
Co-organiser — "AI meets Geometry" Workshop July 13th to 17th, 2026
XXII Escola de Geometria Diferencial (EGD). egd.ufpi.edu.br
Co-organiser — Seminar series on machine learning, Geometry and topology Sep 2026 – present
monica-drechsler.github.io/seminarios-matematica-ml/
Journal Reviewer
Referee for *International Journal of Data Science in the Mathematical Sciences (IJDSMS)* 2026–present
www.worldscientific.com/worldscinet/ijdsms

• **Service to the Community and Public**

Volunteer Mathematics Tutor — "Cursinho Popular Zilda Arns" Jan 2020 – Jan 2021
A non-profit preparatory school supporting underprivileged students in preparing for Brazilian university entrance exams.

• **Professional Development — Invited Programme Participation**

Actions and Intuition: Algebra via AI, Hausdorff Research Institute for Mathematics (HIM) ..... Sep 2026
Thematic Program on AI and Mathematics, Korea Institute for Advanced Study (KIAS) .... Jul–Aug 2026
Mathematics and Machine Learning Program, Harvard CMSA ..... Sep–Nov 2024
London Geometry and Machine Learning (LOGML), Imperial College London ..... Jul 2024

• **Professional Development — Workshops & Conferences Attended**

Inaugural Workshop of the Brazilian Center for Geometry, UNICAMP 2026
Vector Bundles and their Applications in Algebraic Geometry and
Related Topics (Bandoleros), UNICAMP 2025
XLIII National Congress of Applied and Computational Mathematics,
Federal University of Pernambuco 2024
Bridges Meeting on Special Geometries and Stability, Institut d'Études Scientifiques, Cargèse 2024
Hybrid Conference on AI-Mathematics, Rio de Janeiro State University 2024
MATH-AmSud School on Geometry: Group Actions, Symmetries, Moduli and Beyond,
University of Chile 2023
EnCoRi — Encounter on Codes, Lattices and Information Theory, UNICAMP 2023
XV ALGA — Commutative Algebra and Algebraic Geometry, IMPA 2023
XVI Theses, Dissertations and Undergraduate Research Workshop, Computing Institute, UNICAMP 2021
XV Theses, Dissertations and Scientific Undergraduate Research, Computing Institute, UNICAMP .. 2020
I Journey of Undergraduate Research in Codes, Cryptography and Information Theory, UNICAMP .. 2019
Metric and Combinatorial Problems Related to ECC, UNICAMP 2019
XIV Theses, Dissertations and Undergraduate Research Workshop, Computing Institute, UNICAMP 2019
XXVII Congress of Undergraduate Research, UNICAMP 2019
I Latin American Week on Coding and Information, UNICAMP 2018

Languages & Technical Skills

Languages: Portuguese (native), English (advanced), Spanish (advanced), Hebrew (intermediate), French (intermediate).
Programming: C, C++, Python (PyTorch/JAX/TensorFlow), CUDA, Wolfram Mathematica, SageMath, Julia, SQL, Haskell, Linux/Bash, ARM Assembly.
Document Creation: LaTeX, Microsoft Office Suite, Markdown.